SHIFTED HERMITE–BIEHLER KERNEL IDENTITIES, LOCAL MULTIPLICITY JETS, AND TRUNCATED PRIME-POWER MULTIPLIERS FOR RIEMANN'S Ξ-FUNCTION

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Abstract. We establish three exact algebraic and local analytic identities for the transverse Wronskian $W(d,t) = \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2$ of Riemann's Ξ-function, together with explicit circularity and domain obstructions. First, for $E_\alpha(z) = \Xi(z) - \mathrm{i}\alpha\Xi'(z)$ $(\alpha > 0)$, the two-point Nevanlinna–Pick kernel reduces to a Wronskian divided difference with diagonal $K_\alpha(t-\mathrm{i}d, t-\mathrm{i}d) = \frac{\alpha}{\pi}\widetilde{W}(d,t)$, where $\widetilde{W}(d,t) = d^{-1}W(d,t)$; positive semidefiniteness of its Cauchy–Pick matrices is equivalent to the Riemann Hypothesis (Levinson 1974; Lagarias 2005, 2006). Second, at an order-m real zero t_0 of a real-entire function f, the Jensen–Laguerre coefficients $c_{2n-1}(f;t_0) = n\,L_n(f;t_0)$ vanish for $n < m$ while $c_{2m-1}(f;t_0) = \frac{m}{(m!)^2}(f^{(m)}(t_0))^2 > 0$, yielding a local quadratic bowl that separates real Lehmer pairs from off-critical conjugate pairs. Third, we prove the Leibniz–Wronskian multiplier identity, establish $0 \le b_n^{\mathcal{E}}(X) \le 1$ for truncated exponential Euler factors on $\mathrm{Re}(s) > 1$, and isolate the Schwarz-reflection, functional-equation, and strip-residue obstructions on $0 < d \le \frac{1}{2}$.

1. Introduction

Let $\xi(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ denote Riemann's completed zeta function, and let $\Xi(z) := \xi(\frac{1}{2}+\mathrm{i}z)$ be its critical-line rotation [25, 26]. Because $\xi(s) = \xi(1-s)$ and $\overline{\xi(\bar{s})} = \xi(s)$, $\Xi(z)$ is an even real-entire function of order 1 satisfying the Schwarz reflection identity $\Xi^{\#}(z) := \overline{\Xi(\bar{z})} = \Xi(z)$. Writing $z = t - \mathrm{i}d$ with spectral height $t \in \mathbb{R}$ and transverse offset $d = \sigma - \frac{1}{2} \in \mathbb{R}$, we decompose $\Xi(t-\mathrm{i}d) = \xi(\frac{1}{2}+d+\mathrm{i}t) = R(d,t) + \mathrm{i}I(d,t)$ into real-valued harmonic components and consider the transverse Wronskian and its parity-reduced quotient studied in Papers I–III [18, 19, 20]:

$$W(d,t) := \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2 = R(d,t)\,\partial_d R(d,t) + I(d,t)\,\partial_d I(d,t), \qquad \widetilde{W}(d,t) := \frac{W(d,t)}{d}. \tag{1.1}$$

The purpose of the present note is to establish three exact structural identities connecting $\widetilde{W}(d,t)$ to shifted Hermite–Biehler kernels, critical-line multiplicity jets, and holomorphic prime-power multipliers, while demarcating rigorously what each identity proves unconditionally and where equivalence to the Riemann Hypothesis (RH) or strip-analytic obstructions enter:

(1) **Shifted Hermite–Biehler kernels and equivalence to RH (Section 2):** Following Levinson [17], de Branges [6, 7], and Lagarias [14, 15], consider the shifted entire family $E_\alpha(z) := \Xi(z) - \mathrm{i}\alpha\Xi'(z)$ $(\alpha > 0)$. Theorem 2.2 and Proposition 2.3 show unconditionally that the two-point Nevanlinna–Pick kernel $K_\alpha(z_1, z_2)$ collapses to $\frac{\alpha}{\pi}\frac{\Xi(z_1)\overline{\Xi'(z_2)} - \Xi'(z_1)\overline{\Xi(z_2)}}{z_1 - \bar{z}_2}$, so its diagonal $K_\alpha(t-\mathrm{i}d, t-\mathrm{i}d) = \frac{\alpha}{\pi}\widetilde{W}(d,t)$ is an exact scalar multiple of $\widetilde{W}(d,t)$. Proposition 2.4 gives a self-contained proof via Hadamard factorization and Pick's theorem [21] that positive semidefiniteness of the Cauchy–Pick matrices $[K_\alpha(z_j, z_k)]$ (equivalently, the Hermite–Biehler

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property of $E_\alpha^{\#}$ on $\mathbb{C}_+$) holds *if and only if* RH is true, contrasting this $\sigma = \frac{1}{2}$ formulation with de Branges' $\sigma = 1$ shift condition refuted by Conrey and Li [1].

(2) **Jensen–Laguerre multiplicity jets and local quadratic bowls (Section 3):** Using the classical identification $c_{2n-1}(f;t) = n\,L_n(f;t)$ with the Jensen–Laguerre differential polynomials of Jensen [12] and Csordas and Varga [4] (Paper II [19]), Theorem 3.1 proves that for any real-entire function f having a real zero t_0 of multiplicity $m \geq 1$, $c_1(f;t_0) = \cdots = c_{2m-3}(f;t_0) = 0$ and $c_{2m-1}(f;t_0) = \frac{m}{(m!)^2}(f^{(m)}(t_0))^2 > 0$. Theorem 3.2 derives the unified quadratic jet $\widetilde{W}_f(d, \tau_* + \tau) = \frac{1}{2}(f''(\tau_*))^2(\mu + \tau^2 + d^2) + \mathcal{O}((|\mu|^{1/2} + |\tau| + |d|)^3)$ around any non-degenerate critical point $f'(\tau_*) = 0$ with $\mu := -2f(\tau_*)/f''(\tau_*)$, showing that the bowl is strictly positive when $\mu = +\delta^2 > 0$ (real Lehmer pair [16]), vanishes only at the origin when $\mu = 0$ ($m = 2$), and turns strictly negative on the open disk $\tau^2 + d^2 < \delta_0^2$ when $\mu = -\delta_0^2 < 0$ (off-critical conjugate pair, as realized by the Davenport–Heilbronn function [5]).

(3) **Leibniz–Wronskian multipliers, bounded Euler residuals, and strip obstructions (Sections 4–6):** Theorem 4.1 proves the cross-term-free identity $W_{\mathcal{M}\Xi}(d,t) = |\mathcal{M}(z)|^2 W(d,t) + |\Xi(z)|^2 W_{1,\mathcal{M}}(d,t)$ for arbitrary holomorphic multipliers $\mathcal{M}(z)$ and establishes local point-wise zero-exclusion bounds when $W_{\mathcal{M}\Xi}(d_0,t_0) > 0$. Theorem 4.3 proves that the truncated exponential Euler factor $E_X(s) = \exp(-\sum_{2\leq n\leq X} \frac{\Lambda(n)}{\ln n} n^{-s})$ on $\mathrm{Re}(s) > 1$ (cf. Gonek, Hughes, and Keating [9]) produces residual Dirichlet coefficients satisfying $b_n^{\mathcal{E}}(X) = 0$ ($2 \leq n \leq X$), $0 \leq b_n^{\mathcal{E}}(X) \leq 1$ ($n \geq 1$), and $b_n^{\mathcal{E}}(X) = \Lambda(n)/\ln n$ on $X < n < (\lfloor X \rfloor + 1)^2$. Proposition 4.2, Proposition 4.4, and Section 6 then prove explicitly why neither E_X, its Schwarz symmetrization $\mathcal{M}_X^{\mathrm{sym}}$, Fejér Bochner–Weil convolution, nor Gabcke's critical-line Riemann–Siegel bounds [8] yield unconditional off-critical positivity on $0 < d \leq \frac{1}{2}$.

1.1. **Proof overview.** Section 2 expands $|\Xi(z) \mp \mathrm{i}\alpha\Xi'(z)|^2$ via the Cauchy–Riemann equations in (d,t)-coordinates so that the zeroth-order norm $|\Xi|^2$ and quadratic derivative norm $\alpha^2|\Xi'|^2$ cancel identically, leaving $4\alpha W(d,t)$; Hadamard's genus-0 factorization of $\Xi(\sqrt{w})$ together with Pick's theorem then closes the four-way equivalence with RH. Section 3 combines complex Taylor expansion of $f(t_0 - \mathrm{i}d)$ with binomial pairing in $L_n(f;t)$ to isolate the diagonal $j = m$ pure square and expands $f(\tau_* + w)$ around $f'(\tau_*) = 0$ to obtain the μ-parametrized bowl. Section 4 differentiates $\ln|\mathcal{M}\Xi|$ harmonically away from zeros, proves $0 \leq b_n^{\mathcal{E}}(X) \leq 1$ by strong induction on $\sum_{k|n,k>X} \Lambda(k) \leq \ln n$, evaluates the Fejér contour shift across $s = 1$ and off-critical zeros via Poisson–Fourier inversion, and applies the $d = 0$ Leibniz–Laguerre law to $\Xi(t) = -E_0(t)Z(t)$.

1.2. **Notation and conventions.** Throughout the series [18, 19, 20], Riemann's rapid-decay kernel is

$$\Phi(u) := \sum_{n=1}^{\infty} (2\pi^2 n^4 \mathrm{e}^{9u} - 3\pi n^2 \mathrm{e}^{5u}) \exp(-\pi n^2 \mathrm{e}^{4u}), \tag{1.2}$$

which is even and holomorphic on the horizontal strip $|\mathrm{Im}(u)| < \frac{\pi}{8}$ [23, 26]. Under this normalization,

$$\Xi(z) = 4\int_{-\infty}^{\infty} \Phi(u)\,\mathrm{e}^{-2\mathrm{i}zu}\,\mathrm{d}u = 8\int_0^{\infty} \Phi(u)\cos(2zu)\,\mathrm{d}u, \tag{1.3}$$

so for $z = t - \mathrm{i}d$ ($d, t \in \mathbb{R}$), $\Xi(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ with $R(d,t) = 8\int_0^{\infty} \Phi(u)\cosh(2du)\cos(2tu)\,\mathrm{d}u$ and $I(d,t) = 8\int_0^{\infty} \Phi(u)\sinh(2du)\sin(2tu)\,\mathrm{d}u$. Defining the Wigner phase-space marginal integral [30, 18]

$$I_{\mathrm{Wig}}(y,t) := \int_{-\infty}^{\infty} \Phi\left(\frac{x+y}{2}\right)\Phi\left(\frac{x-y}{2}\right)\cos(2tx)\,\mathrm{d}x, \tag{1.4}$$

the canonical integral representations of $W(d,t)$ and $\widetilde{W}(d,t)$ established in Papers I and II are

$$W(d,t) = 16\int_0^\infty y\sinh(2dy)\, I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y, \qquad \widetilde{W}(d,t) = 32\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)\, I_{\mathrm{Wig}}(y,t)\,\mathrm{d}y, \tag{1.5}$$

where $\mathrm{sinc}_{\mathrm{hyp}}(x) := \frac{\sinh x}{x}$ for $x \neq 0$ and $\mathrm{sinc}_{\mathrm{hyp}}(0) := 1$. For any holomorphic function f, we set $f^{\#}(z) := \overline{f(\bar{z})}$; $\mathbb{C}_\pm := \{z \in \mathbb{C} : \pm\,\mathrm{Im}(z) > 0\}$ denote the open upper and lower half-planes.

2. Shifted Hermite–Biehler Kernels and Equivalence to the Riemann Hypothesis

In de Branges' theory of Hilbert spaces of entire functions [6, Ch. 2], an entire function E belongs to the Hermite–Biehler class $\mathcal{HB}(\mathbb{C}_+)$ on the upper half-plane if $E(w) \neq 0$ for $\mathrm{Im}(w) > 0$ and $|E(w)| > |E^{\#}(w)|$ for all $w \in \mathbb{C}_+$. For real-entire functions f ($f^{\#} = f$), Levinson [17] and Lagarias [14, 15] studied the first-order differential shifts $f \pm \mathrm{i}\alpha f'$. Because our transverse coordinate $z = t - \mathrm{i}d$ places $d > 0$ in the lower half-plane $\mathbb{C}_-$, we state both half-plane conventions explicitly.

Definition 2.1 (Shifted family and two-point Nevanlinna–Pick kernel)**.** For any real-entire function $f(z)$ ($f^{\#} = f$) and any parameter $\alpha > 0$, define the entire pair

$$E_\alpha(f;z) := f(z) - \mathrm{i}\alpha\, f'(z), \qquad E_\alpha^{\#}(f;z) := f(z) + \mathrm{i}\alpha\, f'(z). \tag{2.1}$$

When $f = \Xi$, we abbreviate $E_\alpha(z) := E_\alpha(\Xi; z)$. For $z_1, z_2 \in \mathbb{C}$ with $z_1 \neq \bar{z}_2$, define the two-point Hermitian kernel on $\mathbb{C}_-$ ($z_j = t_j - \mathrm{i}d_j$, $d_j > 0$) by

$$K_\alpha(f; z_1, z_2) := \frac{E_\alpha(f;z_1)\overline{E_\alpha(f;z_2)} - E_\alpha^{\#}(f;z_1)\overline{E_\alpha^{\#}(f;z_2)}}{2\pi\mathrm{i}(z_1 - \bar{z}_2)}, \tag{2.2}$$

and write $K_\alpha(z_1,z_2) := K_\alpha(\Xi; z_1, z_2)$. In the conjugate upper half-plane coordinates $w_j := \bar{z}_j = t_j + \mathrm{i}d_j \in \mathbb{C}_+$, since $E_\alpha(\bar{w}) = \overline{E_\alpha^{\#}(w)}$, (2.2) coincides with the upper half-plane de Branges kernel [6, p. 50] for $E_{\mathrm{up},\alpha}(w) := E_\alpha^{\#}(w) = \Xi(w) + \mathrm{i}\alpha\Xi'(w)$:

$$K_\alpha(\bar{w}_1, \bar{w}_2) = \frac{E_\alpha^{\#}(w_2)\overline{E_\alpha^{\#}(w_1)} - E_\alpha(w_2)\overline{E_\alpha(w_1)}}{2\pi\mathrm{i}(\bar{w}_1 - w_2)}. \tag{2.3}$$

Theorem 2.2 (Unconditional diagonal and Schur identities)**.** *Let $\alpha > 0$. At every point $(d,t) \in (\mathbb{R}\setminus\{0\}) \times \mathbb{R}$, the diagonal kernel of $E_\alpha(z) = \Xi(z) - \mathrm{i}\alpha\Xi'(z)$ satisfies the exact algebraic identity*

$$\boxed{K_\alpha(t - \mathrm{i}d, t - \mathrm{i}d) = \frac{|E_\alpha(t-\mathrm{i}d)|^2 - |E_\alpha^{\#}(t-\mathrm{i}d)|^2}{4\pi d} = \frac{\alpha}{\pi}\,\widetilde{W}(d,t),} \tag{2.4}$$

or equivalently $(|E_\alpha^{\#}(t+\mathrm{i}d)|^2 - |E_\alpha(t+\mathrm{i}d)|^2)/(4\pi d) = \frac{\alpha}{\pi}\widetilde{W}(d,t)$ at $w = t + \mathrm{i}d \in \mathbb{C}_+$. Taking $d \to 0$ yields the critical-line limit

$$\lim_{d\to 0} K_\alpha(t - \mathrm{i}d, t - \mathrm{i}d) = \frac{\alpha}{\pi}\,c_1(t) = \frac{\alpha}{\pi}\Big(\big(\Xi'(t)\big)^2 - \Xi(t)\Xi''(t)\Big). \tag{2.5}$$

Furthermore, at every point $z = t - \mathrm{i}d$ ($d > 0$) where $E_\alpha(z) \neq 0$, the meromorphic quotient $\Theta_\alpha(z) := E_\alpha^{\#}(z)/E_\alpha(z)$ satisfies the norm-defect identity

$$1 - |\Theta_\alpha(t - \mathrm{i}d)|^2 = \frac{4\alpha d\,\widetilde{W}(d,t)}{|E_\alpha(t-\mathrm{i}d)|^2}. \tag{2.6}$$

Proof. Write $z = t - \mathrm{i}d$ ($d \neq 0$) and $\Xi(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ with $R, I \in \mathbb{R}$. Holomorphicity in $z = t - \mathrm{i}d$ gives $\Xi'(t-\mathrm{i}d) = \partial_t R + \mathrm{i}\partial_t I$ and $\partial_d \Xi(t - \mathrm{i}d) = -\mathrm{i}\Xi'(t - \mathrm{i}d)$, whence the Cauchy–Riemann equations in (d,t)-coordinates read

$$\partial_d R(d,t) = \partial_t I(d,t), \qquad \partial_d I(d,t) = -\partial_t R(d,t). \tag{2.7}$$

Consequently $-\mathrm{i}\alpha\Xi'(t-\mathrm{i}d) = \alpha\,\partial_d R + \mathrm{i}\alpha\,\partial_d I$, so

$$E_\alpha(t-\mathrm{i}d) = (R+\alpha\,\partial_d R) + \mathrm{i}(I+\alpha\,\partial_d I), \qquad E_\alpha^{\#}(t-\mathrm{i}d) = (R-\alpha\,\partial_d R) + \mathrm{i}(I-\alpha\,\partial_d I). \tag{2.8}$$

Subtracting the squared moduli cancels $R^2+I^2 = |\Xi|^2$ and $\alpha^2((\partial_d R)^2+(\partial_d I)^2) = \alpha^2|\Xi'|^2$ identically:

$$|E_\alpha(t-\mathrm{i}d)|^2 - |E_\alpha^{\#}(t-\mathrm{i}d)|^2 = 4\alpha(R\,\partial_d R + I\,\partial_d I) = 4\alpha\,W(d,t). \tag{2.9}$$

Since $2\pi\mathrm{i}(z-\bar z) = 2\pi\mathrm{i}(-2\mathrm{i}d) = 4\pi d$, dividing (2.9) by $4\pi d$ yields (2.4). Letting $d\to 0$ gives (2.5) by Paper II [19], and dividing (2.9) by $|E_\alpha(t-\mathrm{i}d)|^2$ gives (2.6). □

Proposition 2.3 (Cauchy–Pick divided-difference matrix)**.** *For any points $z_1,\dots,z_M\in\mathbb{C}_-$ ($z_j = t_j - \mathrm{i}d_j$, $d_j>0$) and any $\alpha>0$, the $M\times M$ Hermitian Cauchy–Pick matrix $[\mathbf{G}_\alpha(z_1,\dots,z_M)]_{j,k} := K_\alpha(z_j,z_k)$ satisfies the exact entrywise identity*

$$\big[\mathbf{G}_\alpha(z_1,\dots,z_M)\big]_{j,k} = \frac{\alpha}{\pi}\,\frac{\Xi(z_j)\overline{\Xi'(z_k)} - \Xi'(z_j)\overline{\Xi(z_k)}}{z_j-\bar z_k} \qquad (1\le j,k\le M). \tag{2.10}$$

In particular, $\mathbf{G}_\alpha$ scales linearly with $\alpha>0$, so its eigenvalue inertia is independent of $\alpha>0$.

Proof. See Appendix A. □

Theorem 2.2 and Proposition 2.3 are unconditional algebraic identities valid for any real-entire function. By contrast, whether K_α is a *positive-definite reproducing kernel* is not unconditional: as established by Levinson [17] and Lagarias [14, 15], Hermite–Biehler positivity for $E_\alpha^{\#}(w) = \Xi(w)+\mathrm{i}\alpha\Xi'(w)$ is equivalent to RH. We record a short, self-contained proof of the complete four-way equivalence.

Proposition 2.4 (Equivalence to Hermite–Biehler positivity and RH)**.** *Fix any $\alpha>0$. The following four statements are equivalent:*

(i) *The Riemann Hypothesis holds ($\Xi(z)\neq 0$ for all $z\in\mathbb{C}\setminus\mathbb{R}$).*
(ii) *$\widetilde W(d,t)>0$ for all $(d,t)\in(0,\infty)\times\mathbb{R}$.*
(iii) *$E_\alpha^{\#}(w) = \Xi(w)+\mathrm{i}\alpha\Xi'(w)$ belongs to the Hermite–Biehler class $\mathcal{HB}(\mathbb{C}_+)$ (equivalently, $|\Theta_\alpha(z)|<1$ and $E_\alpha(z)\neq 0$ for all $z\in\mathbb{C}_-$).*
(iv) *For every integer $M\ge 1$ and every tuple $z_1,\dots,z_M\in\mathbb{C}_-$, the Hermitian Cauchy–Pick matrix $\mathbf{G}_\alpha(z_1,\dots,z_M)$ in (2.10) is positive semidefinite.*

Proof. (i) ⇒ (ii): Because $\Xi(z)$ is an even entire function of order 1, $g(w) := \Xi(\sqrt{w})$ is entire of order $1/2<1$ (genus 0), so Hadamard's factorization gives $\Xi(z) = \Xi(0)\prod_{k=1}^\infty(1-z^2/\gamma_k^2)$ with $\sum_{k=1}^\infty|\gamma_k|^{-2}<\infty$ and no exponential prefactor [26, Section 2.12]. Under (i), every $\gamma_k\in(0,\infty)$, so for $z = t-\mathrm{i}d$ with $d>0$ we have $\Xi(z)\neq 0$ and

$$\widetilde W(d,t) = \frac{|\Xi(t-\mathrm{i}d)|^2}{d}\,\mathrm{Im}\Big(\frac{\Xi'(t-\mathrm{i}d)}{\Xi(t-\mathrm{i}d)}\Big) = |\Xi(t-\mathrm{i}d)|^2\sum_{k=1}^\infty\left(\frac{1}{(t-\gamma_k)^2+d^2}+\frac{1}{(t+\gamma_k)^2+d^2}\right)>0, \tag{2.11}$$

which matches Lagarias's positivity property [13].

(ii) ⇒ (iii): If $\widetilde W(d,t)>0$ for all $d>0$, then (2.9) gives $|E_\alpha(t-\mathrm{i}d)|^2 = |E_\alpha^{\#}(t-\mathrm{i}d)|^2 + 4\alpha d\,\widetilde W(d,t) > |E_\alpha^{\#}(t-\mathrm{i}d)|^2\ge 0$ for all $d>0$. Thus $E_\alpha(z)\neq 0$ and $|\Theta_\alpha(z)|<1$ throughout $\mathbb{C}_-$, so $E_\alpha^{\#}(w) = \overline{E_\alpha(\bar w)}\in\mathcal{HB}(\mathbb{C}_+)$.

(iii) ⇒ (iv): Under (iii), $\Theta_\alpha:\mathbb{C}_-\to\mathbb{D}$ is holomorphic and strictly contractive on $\mathbb{C}_-$. By Pick's theorem [21], the Nevanlinna–Pick matrix $[\frac{1-\Theta_\alpha(z_j)\overline{\Theta_\alpha(z_k)}}{2\pi\mathrm{i}(z_j-\bar z_k)}]_{j,k=1}^M$ is positive semidefinite; conjugating by $\mathrm{diag}(E_\alpha(z_1),\dots,E_\alpha(z_M))$ yields $\mathbf{G}_\alpha(z_1,\dots,z_M)\succeq 0$.

(iv) ⇒ (i): Specializing (iv) to $M=1$ at $z_1 = t-\mathrm{i}d$ ($d>0$) gives $K_\alpha(z_1,z_1) = \frac{\alpha}{\pi}\widetilde W(d,t)\ge 0$, so $W(d,t) = \mathrm{Im}(\overline{\Xi(z)}\,\Xi'(z))\ge 0$ on $\mathbb{C}_-$. Suppose toward a contradiction that $\Xi(z_0)=0$ at some $z_0 = t_0-\mathrm{i}d_0\in\mathbb{C}_-$ ($d_0>0$) of multiplicity $k\ge 1$. Writing $\Xi(z_0+re^{\mathrm{i}\theta}) = c_k r^k e^{\mathrm{i}k\theta}+\mathcal{O}(r^{k+1})$ ($c_k\neq 0$)

gives $\operatorname{Im}(\overline{\Xi}\,\Xi') = -k|c_k|^2 r^{2k-1}\sin\theta + \mathcal{O}(r^{2k})$, which is strictly negative for $\theta = \frac{\pi}{2}$ and sufficiently small $r \in (0, d_0)$, contradicting $W \geq 0$ on $\mathbb{C}_-$. By $\Xi(-z) = \Xi(z)$ and $\Xi^{\#} = \Xi$, Ξ has no zeros in $\mathbb{C} \setminus \mathbb{R}$. □

Remark 2.5 (Comparison with de Branges' $\sigma = 1$ shift condition and Conrey–Li). De Branges [7] worked instead with $E(\omega) := \xi(1 - \mathrm{i}\omega)$ anchored at the Euler-product boundary $\sigma = 1$, where $E \in \mathcal{HB}(\mathbb{C}_+)$ holds *unconditionally* because $\zeta(s) \neq 0$ for $\operatorname{Re}(s) > 1$, and sought to deduce RH from a global shift-positivity inequality $\operatorname{Re}\langle F(\omega), \omega^{-1}(F(\omega) - F(0))\rangle_{\mathcal{H}(E)} \geq 0$. Conrey and Li [1] proved that this inner-product inequality fails for $\mathcal{H}(\xi(1 - \mathrm{i}\omega))$. Moving the symmetry axis from $\sigma = 1$ to the critical line $\sigma = \frac{1}{2}$ via $E_\alpha^{\#}(w) = \Xi(w) + \mathrm{i}\alpha\Xi'(w)$ eliminates the extra operator condition of [1], at the exact cost that the Hermite–Biehler axiom $E_\alpha^{\#} \in \mathcal{HB}(\mathbb{C}_+)$ itself ceases to be unconditional and becomes equivalent to RH (Proposition 2.4). Indeed, for the explicit real-entire off-critical zero model $F_{\text{off}}(z) := \Xi(z)(1 + \frac{(z-17.85)^2}{0.25^2})(1 + \frac{(z+17.85)^2}{0.25^2})$, Table 1 and Figure 1(a) confirm that $K_1(F_{\text{off}}; z, z) < 0$, $|\Theta_1(F_{\text{off}}; z)| > 1$, and $\lambda_{\min}(\mathbf{G}_1(F_{\text{off}})) < 0$ near $(d, t) = (0.15, 17.85)$.

3. Jensen–Laguerre Multiplicity Jets and Local Quadratic Bowls

Let $f(z)$ be any real-entire function ($f^{\#} = f$), and set $W_f(d, t) := \frac{1}{2}\partial_d |f(t - \mathrm{i}d)|^2$ and $\widetilde{W}_f(d, t) := d^{-1} W_f(d, t)$. Because $|f(t-\mathrm{i}d)|^2 = f(t-\mathrm{i}d)f(t+\mathrm{i}d)$ is an even real-analytic function of d, expanding in d^2 yields [12, 4, 19]

$$|f(t - \mathrm{i}d)|^2 = \sum_{n=0}^{\infty} L_n(f; t)\, d^{2n}, \qquad \widetilde{W}_f(d, t) = \sum_{n=1}^{\infty} c_{2n-1}(f; t)\, d^{2n-2}, \tag{3.1}$$

where $c_{2n-1}(f; t) = n\, L_n(f; t)$ and $L_n(f; t)$ denotes the n-th classical Jensen–Laguerre differential polynomial [12, 24, 2, 4]:

$$L_n(f; t) := \frac{1}{(2n)!} \sum_{j=0}^{2n} (-1)^{n+j} \binom{2n}{j} f^{(j)}(t)\, f^{(2n-j)}(t), \tag{3.2}$$

$$c_{2n-1}(f; t) = n\, L_n(f; t) = \frac{1}{(2n-1)!} \sum_{j=0}^{2n-1} (-1)^{n+j} \binom{2n-1}{j} f^{(j)}(t)\, f^{(2n-j)}(t). \tag{3.3}$$

Here the second equality in (3.3) follows by pairing j with $2n - j$ ($1 \leq j \leq 2n - 1$) via $\binom{2n-1}{j} + \binom{2n-1}{j-1} = \binom{2n}{j}$ and using $\frac{n}{(2n)!} = \frac{1}{2(2n-1)!}$. In particular, $c_1(f; t) = L_1(f; t) = (f'(t))^2 - f(t)f''(t)$ is the classical Laguerre expression, and $c_3(f; t) = 2L_2(f; t) = \frac{1}{6}(3(f''(t))^2 - 4f'(t)f'''(t) + f(t)f^{(4)}(t))$.

Theorem 3.1 (Order-m critical-line multiplicity jet)**.** *Let $f(z)$ be a real-entire function and suppose $t_0 \in \mathbb{R}$ is a real zero of f of exact multiplicity $m \geq 1$, so that $f(t_0) = f'(t_0) = \cdots = f^{(m-1)}(t_0) = 0$ and $f^{(m)}(t_0) \neq 0$. Then all lower transverse coefficients vanish,*

$$c_1(f; t_0) = c_3(f; t_0) = \cdots = c_{2m-3}(f; t_0) = 0, \tag{3.4}$$

while the first non-vanishing coefficient ($n = m$) is the strictly positive pure square

$$c_{2m-1}(f; t_0) = m\, L_m(f; t_0) = \frac{m}{(m!)^2}\Big(f^{(m)}(t_0)\Big)^2 > 0. \tag{3.5}$$

Consequently, along the transverse line $t = t_0$,

$$\widetilde{W}_f(d, t_0) = \frac{m}{(m!)^2}\Big(f^{(m)}(t_0)\Big)^2 d^{2m-2} + \mathcal{O}(d^{2m}) > 0 \qquad (0 < |d| \ll 1). \tag{3.6}$$

Proof. Since $f^{(r)}(t_0)=0$ for $0\le r\le m-1$, Taylor expansion in d gives $f(t_0-\mathrm{i}d)=\frac{(-\mathrm{i}d)^m}{m!}f^{(m)}(t_0)+\mathcal{O}(d^{m+1})$, and evenness of $|f(t_0-\mathrm{i}d)|^2$ in d yields

$$|f(t_0-\mathrm{i}d)|^2=\frac{d^{2m}}{(m!)^2}\Big(f^{(m)}(t_0)\Big)^2+\mathcal{O}(d^{2m+2}). \tag{3.7}$$

Applying $\frac{1}{2d}\partial_d$ to (3.7) produces (3.6), and uniqueness of the power series (3.1) gives (3.4)–(3.5). Directly from (3.2), $f^{(j)}(t_0)f^{(2n-j)}(t_0)=0$ unless $m\le j\le 2n-m$, which is vacuous for $n\le m-1$ and reduces at $n=m$ to the single positive term $j=m$ with coefficient $\frac{m}{(2m)!}\binom{2m}{m}=\frac{m}{(m!)^2}$. □

To analyze near-coalescence of simple zeros (Lehmer pairs [16, 3]) alongside off-critical conjugate pairs on an equal footing, we expand around an arbitrary non-degenerate real critical point $\tau_*\in\mathbb{R}$.

Theorem 3.2 (Unified local quadratic bowl at a real extremum)**.** *Let $f(z)$ be a real-entire function and let $\tau_*\in\mathbb{R}$ satisfy $f'(\tau_*)=0$ and $f''(\tau_*)\neq 0$. Define the signed discriminant parameter*

$$\mu:=-\frac{2\,f(\tau_*)}{f''(\tau_*)}=\frac{2\,L_1(f;\tau_*)}{\big(f''(\tau_*)\big)^2}\in\mathbb{R}. \tag{3.8}$$

(1) ***Unified local quadratic jet:*** *For local coordinates $\tau:=t-\tau_*$ and transverse offset d, as $|\mu|^{1/2}+|\tau|+|d|\to 0$,*

$$\widetilde{W}_f(d,\tau_*+\tau)=\frac{1}{2}\big(f''(\tau_*)\big)^2(\mu+\tau^2+d^2)+\mathcal{O}\Big(\big(|\mu|^{1/2}+|\tau|+|d|\big)^3\Big). \tag{3.9}$$

In particular:

- Real Lehmer pair ($\mu=+\delta^2>0$, $L_1(f;\tau_*)>0$): *f has two simple real zeros at $\tau_*\pm\delta+\mathcal{O}(\delta^2)$, and $\widetilde{W}_f(d,\tau_*+\tau)=\frac{1}{2}(f''(\tau_*))^2(\delta^2+\tau^2+d^2)+\mathcal{O}((\delta+|\tau|+|d|)^3)$ is a strictly positive sum of three squares near $(0,\tau_*)$.*
- Double real zero ($\mu=0$, $L_1(f;\tau_*)=0$): *$\widetilde{W}_f(d,\tau_*+\tau)=\frac{1}{2}(f''(\tau_*))^2(\tau^2+d^2)+\mathcal{O}((|\tau|+|d|)^3)$ vanishes at $(0,\tau_*)$ and is strictly positive on a punctured neighborhood.*
- Off-critical conjugate pair ($\mu=-\delta_0^2<0$, $L_1(f;\tau_*)<0$): *f has a complex-conjugate zero pair at $\tau_*\pm\mathrm{i}\delta_0+\mathcal{O}(\delta_0^2)$, and $\widetilde{W}_f(d,\tau_*+\tau)=\frac{1}{2}(f''(\tau_*))^2(-\delta_0^2+\tau^2+d^2)+\mathcal{O}((\delta_0+|\tau|+|d|)^3)$ is* strictly negative *inside the open disk $\tau^2+d^2<\delta_0^2+\mathcal{O}(\delta_0^3)$.*

(2) ***Local square-completion away from inflection points:*** *At any $(d,t)\in\mathbb{R}^2$ where $f''(t)\neq 0$,*

$$\widetilde{W}_f(d,t)=\left(f'(t)-\frac{1}{3}d^2\,f'''(t)\right)^2+\frac{d^2}{2}\left(f''(t)+\frac{1}{6}\frac{f(t)f^{(4)}(t)}{f''(t)}\right)^2-f(t)f''(t)+\mathcal{R}_{\ge 5}(f;d,t), \tag{3.10}$$

where $\mathcal{R}_{\ge 5}(f;d,t):=-\frac{d^4}{9}(f'''(t))^2-\frac{d^2}{72}\frac{(f(t)f^{(4)}(t))^2}{(f''(t))^2}+\sum_{n=3}^{\infty}c_{2n-1}(f;t)d^{2n-2}$.

Proof. (1) Write $\rho:=|\mu|^{1/2}+|\tau|+|d|$. Since $f(\tau_*)=-\frac{1}{2}\mu f''(\tau_*)$ and $f'(\tau_*)=0$, Taylor's theorem around τ_* gives $f(\tau_*+\tau)=\frac{1}{2}f''(\tau_*)(\tau^2-\mu)+\mathcal{O}(\rho^3)$, $f'(\tau_*+\tau)=f''(\tau_*)\tau+\mathcal{O}(\rho^2)$, and $f''(\tau_*+\tau)=f''(\tau_*)+\mathcal{O}(\rho)$. Hence

$$\begin{aligned}\big(f'(\tau_*+\tau)\big)^2&=\big(f''(\tau_*)\big)^2\tau^2+\mathcal{O}(\rho^3),\\ -f(\tau_*+\tau)f''(\tau_*+\tau)&=\frac{1}{2}\big(f''(\tau_*)\big)^2(\mu-\tau^2)+\mathcal{O}(\rho^3),\\ c_3(f;\tau_*+\tau)\,d^2&=\frac{1}{2}\big(f''(\tau_*)\big)^2d^2+\mathcal{O}(\rho^3).\end{aligned}$$

Summing these three lines and bounding $\sum_{n=3}^{\infty}c_{2n-1}(f;\tau_*+\tau)d^{2n-2}=\mathcal{O}(d^4)=\mathcal{O}(\rho^4)$ proves (3.9); the root locations $\tau_*\pm\sqrt{\mu}+\mathcal{O}(|\mu|)$ follow from Rouché's theorem applied to $f(\tau_*+w)=\frac{1}{2}f''(\tau_*)(w^2-\mu)+\mathcal{O}(|w|^3)$.

(2) Expanding the two squares in (3.10) recovers $c_1(f;t) + c_3(f;t)d^2$ plus $\frac{d^4}{9}(f'''(t))^2 + \frac{d^2}{72}(f(t)f^{(4)}(t)/f''(t))^2$, which cancels the first two terms of $\mathcal{R}_{\geq 5}(f;d,t)$ to leave (3.1). □

Remark 3.3 (Conditional nature of the quadratic bowl and inflection-point poles). Theorem 3.2 shows that local positivity of $\widetilde{W}_f$ near an extremum τ_* is *not* unconditional: it holds if and only if $\mu \geq 0$, i.e., $L_1(f;\tau_*) = -f(\tau_*)f''(\tau_*) \geq 0$ (no off-critical zeros having just lifted from τ_*). Likewise, in (3.10), the term $-f(t)f''(t)$ changes sign between inflection points and zeros, and the rational completion $(f''(t))^{-2}$ develops cancelling poles at inflection points $f''(t) = 0$, so (3.10) serves strictly as a local jet identity near non-degenerate extrema rather than a global lower bound.

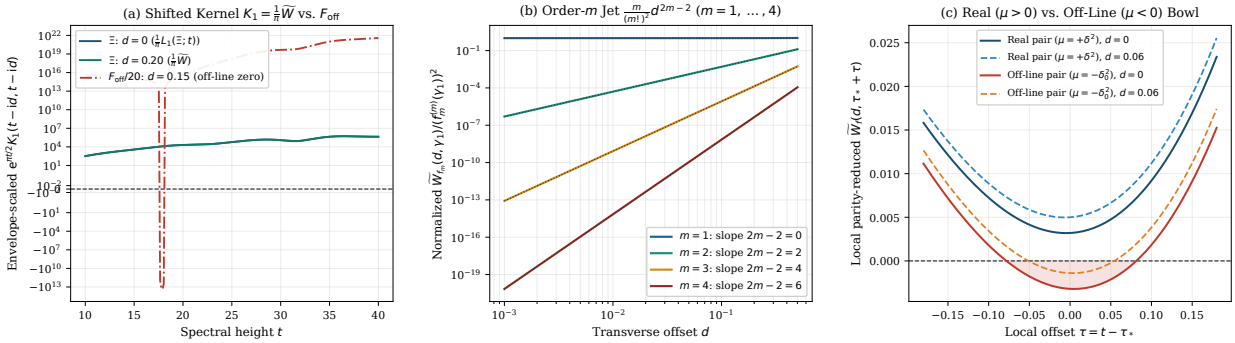


FIGURE 1. **(a)** Envelope-scaled diagonal kernel $e^{\pi t/2}K_1(t-id, t-id) = \frac{1}{\pi}e^{\pi t/2}\widetilde{W}(d,t)$ for Ξ at $d = 0$ and $d = 0.20$ versus the off-critical zero perturbation $F_{\text{off}}/20$ at $d = 0.15$, which dips strictly negative near $t = 17.85$ (Proposition 2.4). **(b)** Log–log verification of the order-m multiplicity jet $\widetilde{W}_{f_m}(d,\gamma_1)/(f_m^{(m)}(\gamma_1))^2 \sim \frac{m}{(m!)^2}d^{2m-2}$ for $m = 1, 2, 3, 4$ (Theorem 3.1). **(c)** Unified local quadratic bowl (Theorem 3.2): strictly positive for a real pair ($\mu = +\delta^2 > 0$) versus negative on $\tau^2 + d^2 < \delta_0^2$ for an off-critical conjugate pair ($\mu = -\delta_0^2 < 0$).

4. Leibniz–Wronskian Multipliers, Truncated Prime-Power Series, and Critical-Line Envelopes

We next examine how $W(d,t)$ transforms under multiplication of $\Xi(z)$ by a holomorphic gauge factor $\mathcal{M}(z)$ and analyze the Dirichlet coefficients of truncated exponential Euler factors.

Theorem 4.1 (Leibniz–Wronskian multiplier identity and local exclusion bounds)**.** *Let $\mathcal{M}(z)$ be holomorphic on a domain $\Omega \subset \mathbb{C}$, and set $\Xi_{\mathcal{M}}(z) := \mathcal{M}(z)\Xi(z)$, $W_{\mathcal{M}}(d,t) := \frac{1}{2}\partial_d|\Xi_{\mathcal{M}}(t-id)|^2$, and $W_{1,\mathcal{M}}(d,t) := \frac{1}{2}\partial_d|\mathcal{M}(t-id)|^2$.*

(1) ***Unscaled and parity-reduced Leibniz–Wronskian laws:*** *For every $z = t - id \in \Omega$,*

$$W_{\mathcal{M}}(d,t) = |\mathcal{M}(t-id)|^2\, W(d,t) + |\Xi(t-id)|^2\, W_{1,\mathcal{M}}(d,t). \tag{4.1}$$

For $d \neq 0$, dividing by d gives $\widetilde{W}_{\mathcal{M}}(d,t) = |\mathcal{M}(z)|^2\widetilde{W}(d,t) + |\Xi(z)|^2 d^{-1}W_{1,\mathcal{M}}(d,t)$. The quotient $\widetilde{W}_{1,\mathcal{M}}(d,t) := d^{-1}W_{1,\mathcal{M}}(d,t)$ extends continuously across $d = 0$ on $\Omega \cap \mathbb{R}$ if and only if $W_{1,\mathcal{M}}(0,t) \equiv 0$ (in particular, whenever $\mathcal{M}^{\#} = \mathcal{M}$), in which case

$$L_1(\Xi_{\mathcal{M}};t) = \big(\mathcal{M}(t)\big)^2 L_1(\Xi;t) + \big(\Xi(t)\big)^2 L_1(\mathcal{M};t). \tag{4.2}$$

(2) ***Point-wise zero-exclusion bounds:*** *If $z_0 = t_0 - \mathrm{i}d_0 \in \Omega$ ($d_0 > 0$) satisfies $W_{\mathcal{M}}(d_0, t_0) > 0$, then $\Xi(z_0) \neq 0$ and $\mathcal{M}(z_0) \neq 0$, with explicit lower bounds*

$$|\Xi(z_0)| \geq \mathcal{B}_{\mathrm{CS}}(d_0, t_0; \mathcal{M}) := \frac{W_{\mathcal{M}}(d_0, t_0)}{|\mathcal{M}(z_0)|\,|\Xi'_{\mathcal{M}}(z_0)|} > 0, \tag{4.3}$$

$$|\Xi(z_0)|^2 + |\Xi'(z_0)|^2 \geq \mathcal{B}_{\mathrm{dist}}(d_0, t_0; \mathcal{M}) := \frac{4\,W_{\mathcal{M}}(d_0, t_0)}{2|W_{1,\mathcal{M}}(d_0, t_0)| + \sqrt{|\mathcal{M}(z_0)|^4 + 4\,W_{1,\mathcal{M}}(d_0, t_0)^2}} > 0. \tag{4.4}$$

Conversely, any parasitic zero $\mathcal{M}(z_0) = 0$ forces $W_{1,\mathcal{M}}(d_0, t_0) = 0$ and $W_{\mathcal{M}}(d_0, t_0) = 0$.

Proof. (1) Away from the isolated zeros of $\mathcal{M}$ and Ξ, $W(d,t) = |\Xi(z)|^2 \operatorname{Im}(\Xi'(z)/\Xi(z))$ and $\Xi'_{\mathcal{M}}/\Xi_{\mathcal{M}} = \Xi'/\Xi + \mathcal{M}'/\mathcal{M}$ yield (4.1), which extends across isolated zeros by real-analyticity. Since $W_{1,\mathcal{M}}(d,t) = W_{1,\mathcal{M}}(0,t) + \widetilde{W}_{1,\mathcal{M}}(0,t)d + \mathcal{O}(d^2)$, $d^{-1}W_{1,\mathcal{M}}(d,t)$ has a finite limit as $d \to 0$ if and only if $W_{1,\mathcal{M}}(0,t) = \operatorname{Im}(\overline{\mathcal{M}(t)}\mathcal{M}'(t)) = 0$; when $\mathcal{M}$ is real-valued on $\Omega \cap \mathbb{R}$, taking $d \to 0$ gives (4.2).

(2) If $\Xi(z_0) = 0$ or $\mathcal{M}(z_0) = 0$, (4.1) gives $W_{\mathcal{M}}(d_0, t_0) = 0$. Cauchy–Schwarz on $W_{\mathcal{M}}(d_0, t_0) = \operatorname{Im}(\overline{\mathcal{M}(z_0)\Xi(z_0)}\,\Xi'_{\mathcal{M}}(z_0))$ gives (4.3). Applying $|W(d_0, t_0)| \leq \frac{1}{2}\eta|\Xi(z_0)|^2 + \frac{1}{2}\eta^{-1}|\Xi'(z_0)|^2$ ($\eta > 0$) to (4.1) and choosing $\eta > 0$ so that $\frac{1}{2}|\mathcal{M}(z_0)|^2\eta^{-1} = \frac{1}{2}|\mathcal{M}(z_0)|^2\eta + |W_{1,\mathcal{M}}(d_0, t_0)|$ solves a quadratic for the common coefficient, yielding (4.4). □

Next, we record the exact Fejér Bochner–Weil convolution formula and its strip residue jump when crossing $s = 1$ and any off-critical zeros.

Proposition 4.2 (Fejér Bochner–Weil prime truncation and strip residues)**.** *For $X > 1$ and $U := \ln X > 0$, define the entire Fejér kernel $K_U(u) := \frac{1}{2\pi U}(\frac{\sin(Uu/2)}{u/2})^2$ (≥ 0 on $\mathbb{R}$), whose Fourier transform $\widehat{K_U}(\xi) = \int_{-\infty}^{\infty} K_U(u)\mathrm{e}^{-\mathrm{i}\xi u}\,\mathrm{d}u = \max(0, 1 - \frac{|\xi|}{U})$ vanishes identically for $|\xi| \geq U$. Set $\mathcal{A}(d,\tau) := \operatorname{Re}(\frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\ln\pi + \frac{1}{2}\psi(\frac{s}{2}))|_{s=\frac{1}{2}+d+\mathrm{i}\tau}$.*

(1) ***Unconditional finite prime sum on $d > \frac{1}{2}$ ($\sigma > 1$):*** *For every $d > \frac{1}{2}$ and $t \in \mathbb{R}$,*

$$\begin{aligned}\mathcal{L}_U(d,t) &:= \int_{-\infty}^{\infty} K_U(u)\,\frac{W(d, t+u)}{|\Xi(t+u-\mathrm{i}d)|^2}\,\mathrm{d}u \\ &= \int_{-\infty}^{\infty} K_U(u)\,\mathcal{A}(d, t+u)\,\mathrm{d}u - \sum_{2 \leq n < X} \frac{\Lambda(n)}{n^{1/2+d}}\left(1 - \frac{\ln n}{U}\right)\cos(t\ln n).\end{aligned} \tag{4.5}$$

(2) ***Pole and zero-residue terms inside $0 < d < \frac{1}{2}$ ($\frac{1}{2} < \sigma < 1$):*** *For $0 < d < \frac{1}{2}$ such that $\zeta(\frac{1}{2} + d + \mathrm{i}\tau) \neq 0$ for all $\tau \in \mathbb{R}$, $\mathcal{L}_U(d,t)$ equals the right-hand side of (4.5) plus the $s = 1$ pole contribution and off-critical zero sum*

$$+2\pi \operatorname{Re} K_U\Big(t + \mathrm{i}(\tfrac{1}{2} - d)\Big) - 2\pi \sum_{d_\rho > d} m_\rho \operatorname{Re} K_U((t - \gamma_\rho) + \mathrm{i}(d_\rho - d)), \tag{4.6}$$

where $\rho = \frac{1}{2} + d_\rho + \mathrm{i}\gamma_\rho$ ranges over any distinct non-trivial zeros of $\zeta(s)$ with $d_\rho > d$ of multiplicity $m_\rho \geq 1$.

Proof. Since $\frac{W(d,\tau)}{|\Xi(\tau - \mathrm{i}d)|^2} = \operatorname{Re}\frac{\xi'(\frac{1}{2}+d+\mathrm{i}\tau)}{\xi(\frac{1}{2}+d+\mathrm{i}\tau)} = \mathcal{A}(d,\tau) + \operatorname{Re}\frac{\zeta'(\frac{1}{2}+d+\mathrm{i}\tau)}{\zeta(\frac{1}{2}+d+\mathrm{i}\tau)}$, for $d > \frac{1}{2}$ the Dirichlet series $-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=2}^{\infty}\Lambda(n)n^{-s}$ converges absolutely and uniformly in τ. Because $K_U \in L^1(\mathbb{R})$ is even and $\mathcal{A}(d, t+u)K_U(u) = \mathcal{O}(|u|^{-2}\ln|u|)$ as $|u| \to \infty$, Fubini's theorem integrates $-\Lambda(n)n^{-1/2-d}\cos((t+u)\ln n)$ term-by-term to $-\Lambda(n)n^{-1/2-d}\widehat{K_U}(\ln n)\cos(t\ln n)$, which vanishes for $n \geq \mathrm{e}^U = X$, proving (4.5). For $0 < d < \frac{1}{2}$, $K_U(w)$ is entire and $\mathcal{O}(|\operatorname{Re} w|^{-2})$ uniformly on horizontal strips, while $\zeta'(s)/\zeta(s) = \mathcal{O}(\ln^2(|\operatorname{Im} s| + 2))$ along horizontal zero-free contours [26, Section 9.6]; shifting $\operatorname{Re}(s) =$

$\frac{1}{2}+d$ rightward to $\sigma_1>1$ crosses the simple pole at $s=1$ (residue -1) and each zero ρ with $d_\rho>d$ (residue $+m_\rho$), yielding (4.6) upon taking real parts. □

We now establish the combinatorial coefficient bounds for truncated exponential Euler factors on $\mathrm{Re}(s)>1$ (compare the truncated Euler-product constructions of Gonek, Hughes, and Keating [9]).

Theorem 4.3 (Bounded residual coefficients of truncated exponential Euler factors)**.** *For any cutoff $X\ge 2$, define the entire zero-free exponential prime-power factor*

$$E_X(s):=\exp\left(-\sum_{2\le k\le X}\frac{\Lambda(k)}{\ln k}\,k^{-s}\right), \tag{4.7}$$

and for $\mathrm{Re}(s)>1$ expand the residual Dirichlet series $\zeta_X(s):=E_X(s)\zeta(s)=\exp(\sum_{k>X}\frac{\Lambda(k)}{\ln k}k^{-s})=1+\sum_{n>X}b_n^{\mathcal{E}}(X)\,n^{-s}$.

(1) ***Unit-interval coefficient bounds and quadratic prime-power window:*** *The coefficients $b_n^{\mathcal{E}}(X)$ satisfy the recursion $b_1^{\mathcal{E}}(X)=1$ and*

$$b_n^{\mathcal{E}}(X)\ln n=\sum_{\substack{k\mid n\\ k>X}}b_{n/k}^{\mathcal{E}}(X)\,\Lambda(k)\qquad(n\ge 2), \tag{4.8}$$

together with the exact bounds

$$b_n^{\mathcal{E}}(X)=0\ \ (2\le n\le X),\quad 0\le b_n^{\mathcal{E}}(X)\le 1\ \ (n\ge 1),\quad b_n^{\mathcal{E}}(X)=\frac{\Lambda(n)}{\ln n}\ \ (X<n<(\lfloor X\rfloor+1)^2). \tag{4.9}$$

(2) ***Residual theta-tail monotonicity:*** *For $\psi_X(x):=\sum_{n=2}^{\infty}b_n^{\mathcal{E}}(X)\mathrm{e}^{-\pi n^2x}$ $(x>0)$ and every integer $r\ge 0$,*

$$0\le(-1)^r\psi_X^{(r)}(x)\le\sum_{n=\lfloor X\rfloor+1}^{\infty}(\pi n^2)^r\mathrm{e}^{-\pi n^2x}. \tag{4.10}$$

Proof. See Appendix B. □

Finally, to relate $c_1(t)=L_1(\Xi;t)$ on the critical line $d=0$ accurately to Hardy's function $Z(t):=\mathrm{e}^{\mathrm{i}\vartheta(t)}\zeta(\frac{1}{2}+\mathrm{i}t)$ (where $\vartheta(t):=\mathrm{Im}\ln\Gamma(\frac{1}{4}+\mathrm{i}\frac{t}{2})-\frac{t}{2}\ln\pi$), we record the exact Stirling envelope factorization.

Proposition 4.4 (Exact critical-line envelope identity between $\Xi(t)$ and $Z(t)$)**.** *For all $t\in\mathbb{R}$, $\Xi(t)=-E_0(t)\,Z(t)$, where the positive even real-analytic envelope*

$$E_0(t):=\frac{1}{2}\left(t^2+\frac{1}{4}\right)\pi^{-1/4}\left|\Gamma\left(\frac{1}{4}+\mathrm{i}\frac{t}{2}\right)\right|=\frac{t^2}{\sqrt{2}}\left(\frac{2\pi}{t}\right)^{1/4}\mathrm{e}^{-\pi t/4}\Big(1+\mathcal{O}(t^{-1})\Big)\qquad(t\to+\infty) \tag{4.11}$$

satisfies the exact Laguerre decomposition

$$c_1(t)=L_1(\Xi;t)=E_0(t)^2\,L_1(Z;t)+Z(t)^2\,L_1(E_0;t), \tag{4.12}$$

with $L_1(E_0;t)>0$ for all $|t|\ge 1$.

Proof. Since $(\frac{1}{2}+\mathrm{i}t)(-\frac{1}{2}+\mathrm{i}t)=-(t^2+\frac{1}{4})$ and $\Gamma(\frac{1}{4}+\mathrm{i}\frac{t}{2})\pi^{-\mathrm{i}t/2}=|\Gamma(\frac{1}{4}+\mathrm{i}\frac{t}{2})|\mathrm{e}^{\mathrm{i}\vartheta(t)}$, the definition of $\xi(\frac{1}{2}+\mathrm{i}t)$ gives $\Xi(t)=-E_0(t)Z(t)$, and Stirling's formula $|\Gamma(\frac{1}{4}+\mathrm{i}\frac{t}{2})|=\sqrt{2\pi}(t/2)^{-1/4}\mathrm{e}^{-\pi t/4}(1+\mathcal{O}(t^{-1}))$ gives the asymptotic in (4.11) (note the polynomial factor t^2 and minus sign separating $\Xi(t)$ from $\mathrm{e}^{-\pi t/4}t^{-1/4}Z(t)$). Applying (4.2) to $\Xi(t)=(-E_0(t))Z(t)$ gives (4.12). Differentiating $\ln E_0(t)=-\ln 2+\ln(t^2+\frac{1}{4})-\frac{1}{4}\ln\pi+\mathrm{Re}\ln\Gamma(\frac{1}{4}+\mathrm{i}\frac{t}{2})$ twice yields

$$-\frac{\mathrm{d}^2}{\mathrm{d}t^2}\ln E_0(t)=\frac{2(t^2-\frac{1}{4})}{(t^2+\frac{1}{4})^2}+\frac{1}{4}\,\mathrm{Re}\,\psi'\left(\frac{1}{4}+\mathrm{i}\frac{t}{2}\right).$$

For $|t| \geq 1$, the first term is $\geq \frac{2(3/4)}{(5/4)^2} = \frac{24}{25} = 0.96$, while by Binet's formula $|\psi'(w) - w^{-1} - \frac{1}{2}w^{-2}| \leq \frac{1}{6}|w|^{-3}$ on $\mathrm{Re}(w) = \frac{1}{4}$, $\frac{1}{4}|\mathrm{Re}\,\psi'(\frac{1}{4} + \mathrm{i}\frac{t}{2})| < 0.70$, so $L_1(E_0; t) = -E_0(t)^2 \frac{\mathrm{d}^2}{\mathrm{d}t^2} \ln E_0(t) > 0$ for all $|t| \geq 1$. □

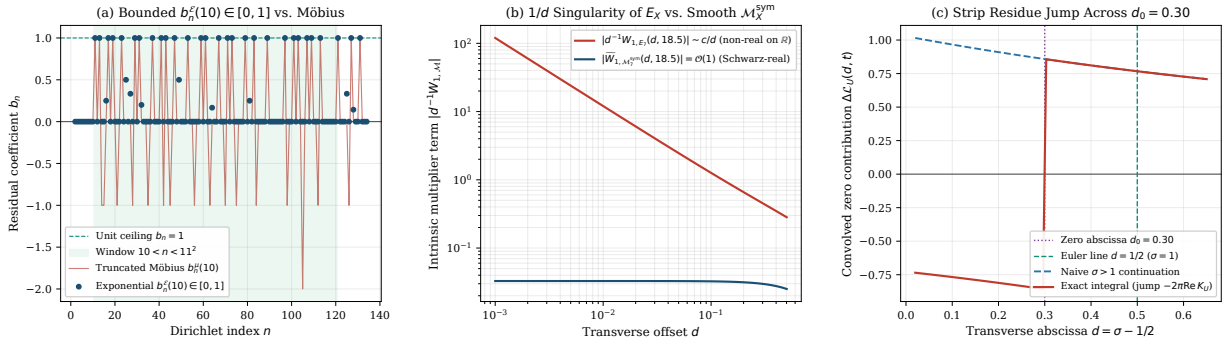


FIGURE 2. **(a)** Residual Dirichlet coefficients $b_n^{\mathcal{E}}(10) \in [0,1]$ of $\zeta_{10}(s) = E_{10}(s)\zeta(s)$ showing prime-power purity on $10 < n < 11^2 = 121$ versus the > 1 overshoot of truncated Möbius convolution (Theorem 4.3). **(b)** Log–log transverse scan at $t = 18.5$: unsymmetrized $|d^{-1}W_{1,E_7}(d, 18.5)| \sim c/d$ diverges as $d \to 0^+$ due to broken Schwarz reflection on $\mathbb{R}$, whereas conjugate-symmetrized $|\widetilde{W}_{1,\mathcal{M}_7^{\mathrm{sym}}}(d, 18.5)|$ remains bounded (Section 6). **(c)** Fejér Bochner–Weil contour shift across an off-critical zero at $d_0 = 0.30$: exact match for $d > d_0$ versus the residue jump $-2\pi\,\mathrm{Re}\,K_U((t - \gamma_0) + \mathrm{i}(d_0 - d))$ on $0 < d < d_0$ (Proposition 4.2).

5. High-Precision Numerical and Interval Verification

All analytical identities, counterexamples, and corrected critical-line constants in Sections 2–4 are verified in the standalone script `scripts/verify_paper4_debranges_lehmer_cascade.py` using 50-digit `mpmath`, 55-digit directed-rounding interval arithmetic (`mpmath.iv`), and 560-digit local Taylor evaluation.

(1) **Canonical Wigner and Cauchy–Pick audit (Table 1):** Across representative strip points, the canonical Wigner integral (1.5) matches $\frac{1}{2}\partial_d|\Xi(t - \mathrm{i}d)|^2$ to relative error $\leq 1.05 \times 10^{-57}$ at 65-digit working precision. For $\alpha \in \{0.25, 1.0, 2.5\}$, the diagonal identity (2.4), Schur defect (2.6), and 5×5 divided-difference matrix (2.10) hold to $\leq 1.80 \times 10^{-50}$. Using $N_u = 160$ composite trapezoidal intervals on $u \in [0, 1.55]$ in `mpmath.iv`—where analyticity of $\Phi(u)$ on $|\mathrm{Im}(u)| \leq 0.35 < \frac{\pi}{8}$ yields a proven Trefethen–Weideman discretization error $< 10^{-65}$ for $d \leq 0.50, t \leq 42$ (Paper II [19, Lemma A.1(c)])—we obtain rigorous enclosures of $\mathrm{e}^{\pi t/2}K_1(t - \mathrm{i}d, t - \mathrm{i}d) > 0$ of width $\leq 2.01 \times 10^{-39}$. Note that finite-box verification at low height is purely illustrative, since RH is already rigorously established up to $t \leq 3 \times 10^{12}$ by Gram [10], Titchmarsh [27], and Platt and Trudgian [22].

(2) **True critical-line extrema τ_1, τ_2 and 550-digit coalescence jet (Table 2):** Because $\Xi(t) = -E_0(t)Z(t)$ contains the exponential factor $\mathrm{e}^{-\pi t/4}$, the roots of $\Xi'(t) = 0$ are shifted substantially downward relative to the extrema of Hardy's Z-function: on (γ_1, γ_2) and (γ_2, γ_3), bracketed root-finding gives

$$\tau_1 = 15.585708589829342345\ldots, \qquad \tau_2 = 22.097977280400902098\ldots, \tag{5.1}$$

whereas the first positive root of $Z'(t) = 0$ lies at $t_{Z',1} = 17.882582076936682719\ldots$ (and the zeroth Gram point $\vartheta(g_0) = 0$ at $g_0 = 17.8455995405\ldots$). Anchoring the local deformation

$f_\delta(z) := \Xi(z) - \Xi(\tau_1) - \frac{1}{2}\Xi''(\tau_1)\delta^2$ at the true extremum τ_1 at 560-digit precision confirms that as $\Delta\gamma = 2\delta + \mathcal{O}(\delta^3)$ shrinks from 10^{-1} to 10^{-250}, $c_1(f_\delta;\tau_1) = \frac{1}{2}(\Xi''(\tau_1))^2\delta^2 \to 0^+$ quadratically while $c_3(f_\delta;\tau_1)$ converges to $\frac{1}{2}(\Xi''(\tau_1))^2 = 1.00779672376\cdots \times 10^{-7} > 0$. Evaluating Lehmer's historical pair $(\gamma_{6709}, \gamma_{6710})$ ($\Delta\gamma \approx 0.03769850$, $\tau_* \approx 7005.08171569$) similarly confirms $\mu = +3.55140363 \times 10^{-4} > 0$, $e^{\pi\tau_*/2}c_1(\tau_*) \approx +3.1959 \times 10^{12} > 0$, and $e^{\pi\tau_*/2}c_3(\tau_*) \approx +9.0067 \times 10^{15} > 0$.

TABLE 1. Verification of the diagonal kernel identity $K_1(z,z) = \frac{1}{\pi}\widetilde{W}(d,t)$ (Theorem 2.2) via 55-digit `mpmath.iv` enclosures ($N_u = 160$, proven strip error $< 10^{-65}$), contrasted with the off-critical zero perturbation $F_{\text{off}}(z)$ at $z = 17.85 - 0.15\text{i}$.

Function	Point (d,t)	$e^{\pi t/2}K_1(z,z)$	Interval Width	$\lvert\Theta_1(z)\rvert$
$\Xi(z)$	$(0.05,\ 14.13472514)$	$+2.67305185 \times 10^{3}$	1.16×10^{-48}	$0.985410 < 1$
$\Xi(z)$	$(0.20,\ 15.58570859)$	$+4.94383597 \times 10^{3}$	2.06×10^{-48}	$0.761529 < 1$
$\Xi(z)$	$(0.35,\ 21.02203964)$	$+2.27454159 \times 10^{4}$	7.53×10^{-46}	$0.684912 < 1$
$\Xi(z)$	$(0.50,\ 25.01085758)$	$+6.37572008 \times 10^{4}$	2.37×10^{-44}	$0.619404 < 1$
$\Xi(z)$	$(0.25,\ 37.58617816)$	$+5.02658084 \times 10^{5}$	2.01×10^{-39}	$0.793118 < 1$
$F_{\text{off}}(z)$	$(0.15,\ 17.85000000)$	$-4.48208012 \times 10^{14}$	—	$1.306865 > 1$

TABLE 2. 550-digit local coalescence audit for $f_\delta(z) = \Xi(z) - \Xi(\tau_1) - \frac{1}{2}\Xi''(\tau_1)\delta^2$ anchored at the true first positive root $\tau_1 = 15.5857085898293423\ldots$ of $\Xi'(t) = 0$, with double-zero floor $c_{3,\text{floor}} := \frac{1}{2}(\Xi''(\tau_1))^2 = 1.00779672376 \times 10^{-7}$.

Target 2δ	True Zero Gap $\Delta\gamma$	$c_1(f_\delta;\tau_1)$	$c_3(f_\delta;\tau_1)$	$\lvert c_3 - c_{3,\text{floor}}\rvert / c_{3,\text{floor}}$
10^{-1}	$1.0001473951 \times 10^{-1}$	$2.51949181 \times 10^{-10}$	$1.00748153 \times 10^{-7}$	3.12759×10^{-4}
10^{-3}	$1.0000000147 \times 10^{-3}$	$2.51949181 \times 10^{-14}$	$1.00779669 \times 10^{-7}$	3.12759×10^{-8}
10^{-6}	$1.0000000000 \times 10^{-6}$	$2.51949181 \times 10^{-20}$	$1.00779672 \times 10^{-7}$	3.12759×10^{-14}
10^{-12}	$1.0000000000 \times 10^{-12}$	$2.51949181 \times 10^{-32}$	$1.00779672 \times 10^{-7}$	3.12759×10^{-26}
10^{-30}	$1.0000000000 \times 10^{-30}$	$2.51949181 \times 10^{-68}$	$1.00779672 \times 10^{-7}$	3.12759×10^{-62}
10^{-100}	$1.0000000000 \times 10^{-100}$	$2.51949181 \times 10^{-208}$	$1.00779672 \times 10^{-7}$	3.12759×10^{-202}
10^{-250}	$1.0000000000 \times 10^{-250}$	$2.51949181 \times 10^{-508}$	$1.00779672 \times 10^{-7}$	3.12759×10^{-502}

6. Scope, Limitations, and Explicit Obstructions

We summarize the structural obstructions explaining why the identities in Sections 2–4 do not yield an unconditional proof of off-critical zero-freeness:

(1) **Hermite–Biehler circularity (Proposition 2.4):** Because $K_\alpha(t - \mathrm{id}, t - \mathrm{id}) = \frac{\alpha}{\pi}\widetilde{W}(d,t)$ is an exact scalar multiple of $\widetilde{W}(d,t)$, invoking de Branges' reproducing-kernel Hilbert space $\mathcal{H}(E_\alpha^\#)$ requires knowing $E_\alpha^\# \in \mathcal{HB}(\mathbb{C}_+)$ beforehand, which is equivalent to RH [17, 14, 15].

(2) **Davenport–Heilbronn sign reversal in the quadratic bowl (Theorem 3.2):** The order-m cascade (3.5) and positive bowl $\frac{1}{2}(f''(\tau_*))^2(\delta^2+\tau^2+d^2)$ hold conditionally on the zeros being *real* ($\mu \geq 0$). Any real-entire function possessing off-critical conjugate zeros near $\mathbb{R}$—such as the Davenport–Heilbronn function [5, 26]—exhibits critical-line extrema with $\mu = -\delta_0^2 < 0$ ($L_1 < 0$), reversing the bowl floor to $-\frac{1}{2}(f''(\tau_*))^2\delta_0^2 < 0$.

(3) **Broken Schwarz reflection of E_X versus broken symmetry of $\mathcal{M}_X^{\mathrm{sym}}$:** Set $\mathcal{M}_X(z) := E_X(\frac{1}{2}+\mathrm{i}z)$ with E_X from (4.7). Since $\mathcal{M}_X'(z)/\mathcal{M}_X(z) = \mathrm{i}\sum_{2\le n\le X}\Lambda(n)n^{-1/2-\mathrm{i}z}$, on the critical line $d=0$ one has

$$W_{1,\mathcal{M}_X}(0,t) = \left|E_X(\tfrac{1}{2}+\mathrm{i}t)\right|^2 \sum_{2\le n\le X}\frac{\Lambda(n)}{\sqrt{n}}\cos(t\ln n) \not\equiv 0, \tag{6.1}$$

so $d^{-1}W_{1,\mathcal{M}_X}(d,t)$ diverges like $d^{-1}W_{1,\mathcal{M}_X}(0,t)$ as $d\to 0^+$ (Figure 2(b)). Symmetrizing via $\mathcal{M}_X^{\mathrm{sym}}(z) := E_X(\frac{1}{2}+\mathrm{i}z)E_X(\frac{1}{2}-\mathrm{i}z) = \exp(-2\sum_{2\le n\le X}\frac{\Lambda(n)}{n^{1/2}\ln n}\cos(z\ln n))$ restores $\mathcal{M}^{\#}=\mathcal{M}$ and yields the smooth parity-reduced Wronskian

$$\widetilde{W}_{1,\mathcal{M}_X^{\mathrm{sym}}}(d,t) = -2\left|\mathcal{M}_X^{\mathrm{sym}}(t-\mathrm{i}d)\right|^2 \sum_{2\le n\le X}\frac{\Lambda(n)\ln n}{\sqrt{n}}\,\mathrm{sinc}_{\mathrm{hyp}}(d\ln n)\cos(t\ln n). \tag{6.2}$$

However, two simultaneous obstructions arise: first, $E_X(s)\neq E_X(1-s)$ breaks the functional equation of $\xi(s)$, so neither $E_X(s)\xi(s)$ nor $\mathcal{M}_X^{\mathrm{sym}}(z)\Xi(z)$ admits a symmetric Riemann–Siegel main sum with coefficients $b_n^{\mathcal{E}}(X)$; second, by Kronecker's linear independence of $\{\ln p\}$, (6.2) has mean zero in t and oscillates between positive and negative values, subtracting from $|\mathcal{M}_X^{\mathrm{sym}}|^2\widetilde{W}(d,t)$ in (4.1) whenever primes align in phase.

(4) **Strip residue barrier and Gabcke domain limits:** In Proposition 4.2, the prime-truncated identity (4.5) holds unconditionally only on $d>\frac{1}{2}$ ($\sigma>1$, cf. Hinkkanen [11]); inside $0<d<\frac{1}{2}$, omitting the zero-residue sum in (4.6) assumes a priori that no zeros lie to the right of $\frac{1}{2}+d$, while Voronin's universality theorem [29] governs the arbitrary phase behavior of $\zeta'(s)/\zeta(s)$ on $(\frac{1}{2},1)$. Likewise, Gabcke's explicit Riemann–Siegel remainder bounds [8] apply strictly to unmodified $Z(t)$ ($b_n\equiv 1$, $d=0$, $t\ge 200$) and cannot be extrapolated off the critical line ($d>0$) or to sieved weights $b_n^{\mathcal{E}}(X)$; even on $d=0$, extreme Skewes-type phase alignments allow the main-sum approximation of $L_1(Z;t)$ to dip below remainder thresholds near close zero pairs, requiring Turing zero-counting [28, 22] rather than main-sum dominance alone.

Appendix A. Derivation of the Cauchy–Pick Divided-Difference Matrix

Proof of Proposition 2.3. Let $z_j, z_k\in\mathbb{C}_-$. Expanding $E_\alpha(z_j)\overline{E_\alpha(z_k)}$ and $E_\alpha^{\#}(z_j)\overline{E_\alpha^{\#}(z_k)}$ from (2.1) gives

$$E_\alpha(z_j)\overline{E_\alpha(z_k)} = \Xi(z_j)\overline{\Xi(z_k)} + \alpha^2\Xi'(z_j)\overline{\Xi'(z_k)} + \mathrm{i}\alpha\Big(\Xi(z_j)\overline{\Xi'(z_k)} - \Xi'(z_j)\overline{\Xi(z_k)}\Big),$$
$$E_\alpha^{\#}(z_j)\overline{E_\alpha^{\#}(z_k)} = \Xi(z_j)\overline{\Xi(z_k)} + \alpha^2\Xi'(z_j)\overline{\Xi'(z_k)} - \mathrm{i}\alpha\Big(\Xi(z_j)\overline{\Xi'(z_k)} - \Xi'(z_j)\overline{\Xi(z_k)}\Big).$$

Subtracting the second line from the first cancels the zeroth-order and α^2 products for all pairs (j,k), leaving $2\mathrm{i}\alpha(\Xi(z_j)\overline{\Xi'(z_k)} - \Xi'(z_j)\overline{\Xi(z_k)})$. Dividing by $2\pi\mathrm{i}(z_j-\bar{z}_k)\neq 0$ yields (2.10). □

Appendix B. Proof of the Truncated Exponential Euler Coefficient Bounds

Proof of Theorem 4.3. (1) For $\mathrm{Re}(s)>1$, $\ln\zeta_X(s) = \sum_{k>X}\frac{\Lambda(k)}{\ln k}k^{-s}$ converges absolutely. Differentiating $\zeta_X(s)=\sum_{n=1}^{\infty}b_n^{\mathcal{E}}(X)n^{-s}$ gives $-\zeta_X'(s) = \zeta_X(s)(-\ln\zeta_X(s))'$, i.e.,

$$\sum_{n=2}^{\infty} b_n^{\mathcal{E}}(X)(\ln n)\,n^{-s} = \left(\sum_{m=1}^{\infty} b_m^{\mathcal{E}}(X)\,m^{-s}\right)\left(\sum_{k>X}\Lambda(k)\,k^{-s}\right).$$

Equating coefficients of n^{-s} yields (4.8) with $b_1^{\mathcal{E}}(X)=1$. For $2\le n\le X$, the divisor set $\{k\mid n : k>X\}$ is empty, so $b_n^{\mathcal{E}}(X)=0$. Since $\Lambda(k)\ge 0$, strong induction on (4.8) gives $b_n^{\mathcal{E}}(X)\ge 0$ for

all $n \geq 1$. Assuming $b_m^{\mathcal{E}}(X) \leq 1$ for all $1 \leq m < n$, (4.8) and the fundamental divisor identity $\sum_{k|n} \Lambda(k) = \ln n$ give

$$b_n^{\mathcal{E}}(X) \ln n = \sum_{\substack{k|n \\ k>X}} b_{n/k}^{\mathcal{E}}(X)\, \Lambda(k) \leq \sum_{\substack{k|n \\ k>X}} \Lambda(k) \leq \sum_{k|n} \Lambda(k) = \ln n,$$

proving $b_n^{\mathcal{E}}(X) \leq 1$ for all $n \geq 1$. Finally, let $X < n < (\lfloor X \rfloor + 1)^2$. For any $k \mid n$ with $k > X$, we have $k \geq \lfloor X \rfloor + 1$, so $n/k < \lfloor X \rfloor + 1$, whence $1 \leq n/k \leq \lfloor X \rfloor \leq X$. Because $b_m^{\mathcal{E}}(X) = 0$ for $2 \leq m \leq X$, only $n/k = 1$ ($k = n$) contributes to (4.8), giving $b_n^{\mathcal{E}}(X) \ln n = b_1^{\mathcal{E}}(X)\Lambda(n) = \Lambda(n)$.

(2) Differentiating $\psi_X(x)$ r times term-by-term for $x > 0$ gives

$$(-1)^r \psi_X^{(r)}(x) = \sum_{n=\lfloor X \rfloor + 1}^{\infty} b_n^{\mathcal{E}}(X)(\pi n^2)^r \mathrm{e}^{-\pi n^2 x},$$

and applying $0 \leq b_n^{\mathcal{E}}(X) \leq 1$ yields (4.10). □

Use of AI and Computational Tools

Gemini (Google DeepMind) was used interactively to assist with symbolic verification of the divided-difference and Taylor jet identities, literature cross-checks, LaTeX manuscript preparation, and implementation of the Python/`mpmath`/`mpmath.iv` verification script (`scripts/verify_paper4_debranges_lehmer_cascade.py`). All mathematical proofs, domain conditions, counterexamples, citations, and numerical outputs were independently checked and verified.

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